Executive Summary

Aerospace Structural Design Dataset | Synthetic Data Generation & Validation

OBJECTIVE

Generate high-fidelity synthetic data from a 300-row aerospace structural design dataset, preserving statistical distributions, feature interdependencies, input-to-output coherence, and class balance across 4 categorical target variables.

RECOMMENDED MODEL: KDE (Kernel Density Estimation)

Composite Score	**0.9965 / 1.0**	Weighted aggregate of all quality dimensions
MMD	**0.00027**	Near-zero = distributions statistically indistinguishable
Discriminator AUC	**0.478**	Near-random (0.50) = classifier cannot tell real from synthetic
TSTR Accuracy	**1.000**	Model trained on synthetic achieves perfect accuracy on real data
Schema Compliance	**22/22 cols, 0 nulls, 0 OOR**	Full schema match, zero out-of-range values
Output Size	**5,000 rows**	16.7x the original dataset

CLIENT CRITERIA CHECKLIST

[OK]	**Statistical Fidelity**	KS test 100% pass rate, KL divergence 0.004, all distributions shape-matched
[OK]	**Feature Interdependencies**	Correlation Frobenius norm 0.20 (convergent at n=5000), structure preserved
[OK]	**Input-to-Output Coherence**	SHAP top-5 feature overlap 3/5, 12 domain physics rules tested, TSTR=1.0
[OK]	**Class Balance & Edge Cases**	Chi-square p > 0.83 all targets, 27 edge scenarios at 95.9% Tier-1 pass
[OK]	**Rare Case Realism**	Isolation Forest + LOF + NN-distance validated; mean realism score 90.1%
[OK]	**Overall Similarity**	MMD, TSTR/TRTS, Discriminator AUC all validated with 30-seed bootstrap CIs

DEEPER VALIDATIONS PERFORMED

I/P to O/P Correlations	Correlation matrix, grouped stats by target class, conditional distributions
Feature Structure	Per-column histograms vs reference, KS test, KL divergence, joint pair plots
Class Balance Tests	Class frequency counts, chi-square tests, joint 4-way target distribution
Rare Case Realism	Isolation Forest, LOF, nearest-neighbour distance to reference rows
Overall Similarity	Train-on-synthetic/test-on-real (TSTR), MMD, discriminator AUC with bootstrap CIs

MODEL BENCHMARKING (9 Models, 92 Configurations)

Model	Tier	Composite	MMD	Disc. AUC	TSTR	Verdict
KDE	Production	0.9965	0.00027	0.478	1.000	Best overall - recommended
GMM	Production	0.8535	0.0007	0.451	0.460	Excellent discriminator score
Copula	Production	0.8309	0.0013	0.518	0.347	Best indistinguishability (96.5%)
TabuDiff	Research	0.7890	0.0207	0.722	0.313	Best deep-learning model
MaskedPredictor	Research	0.5250	0.0276	0.976	0.303	High variance, conditional learning
TVAE/CTGAN	Evaluated	0.610	> 0.4	~1.0	0.300	Class collapse - not recommended

DELIVERABLE CONTENTS

- 7 PDF Reports (EDA, Model Comparison, Feature Interdependency, I/O Coherence, Class Balance, Similarity, Business Narrative)
- 294 Catalogued Visualizations across 11 categories with searchable graph_catalog.csv
- 9 Best-per-model synthetic datasets + Top 10 leaderboard + 92 experimental archives
- Full validation pipeline: Anomaly detection, class balancing, edge cases, discriminator tests, MMD, PCA/t-SNE

RECOMMENDED READING ORDER

1. This Summary 2. Model Comparison Report 3. Similarity & Innovation Report 4. Deep Data Analysis Report 5. Remaining reports as needed